

Arpan Pathak

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Visa: Requires H1B Transfer sponsorship

PERSONAL PROJECTS

Driving-CivicSense (Edge AI Vision) | github.com/arpanpathak/driving-civicsense-vision-model | 2026

- On-device AI perception system built in **Rust** for intersection discipline and lane awareness — runs entirely on-device (**< 50ms inference latency**, no cloud dependency) for aftermarket dashcam and AR-glasses form factors.
- Fine-tunes **YOLOv8/v11** and custom **CNN** architectures; applies model pruning and quantization to fit tight memory and power budgets on low-powered single-board computers (Raspberry Pi 5).
- Trains on cloud GPU infrastructure with synthetic data augmentation; planning to release a public driving-behavior dataset on **Hugging Face**.

SUMMARY

Systems Software Engineer with 8+ years architecting performance-critical backend systems, distributed infrastructure, and real-time **ML inference** pipelines at **Amazon, Microsoft, and Oracle**. Deep expertise in **Rust, C++, Go, and Python** — lock-free concurrency, memory safety, low-latency streaming (**Kafka**), and on-device inference. Proven track record shipping high-throughput, resource-efficient systems at global scale; focused on systems programming for consumer electronics, where performance, memory budgets, and real-time responsiveness are the product.

EXPERIENCE

Software Developer II Microsoft — Redmond, WA June 2025 – Present

- **C++ to Rust Migration:** Spearheaded migration of the Information Protection service from **C++ to Rust**, shifting memory safety to compile-time. Eliminated use-after-free and buffer-overflow CVEs in the audited path while maintaining full performance parity — a critical win for DORA compliance.
- **AI-Powered Policy Engine:** Built an AI-driven engine that auto-generates Cilium network policies and scans for misconfigurations, reducing manual policy review cycles from **4 hours** to near-real-time with zero policy drift across sovereign regions.
- **eBPF & Kernel-Level Networking:** Designed and deployed eBPF-based network policies using **Cilium** for Microsoft Purview's sovereign cloud regions. Authored custom **Kubernetes Operators** (Kubebuilder) to reconcile compliance states across AKS clusters.
- **Developer Workflows:** Established CI/CD pipelines, containerized microservices, and automated validation frameworks ensuring reliable multi-region releases.
- Infra: **Rust, C++, eBPF, Cilium, Kubernetes (AKS), Kubebuilder, Docker, Linux**

Senior Software Engineer Oracle Cloud Infrastructure — Seattle, WA Oct 2024 – Mar 2025

- **High-Performance Engine:** Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to reduce contention in high-concurrency read-write paths, improving transaction throughput by **45,000 ops/s**.
- **Infrastructure-as-Code SDK:** Engineered a Terraform provider (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control plane infrastructure with projected **\$1.2M/month cost savings**.
- **Security & Key Management:** Integrated OCI Security Vault across control plane services for secure secret management and automated key rotation protocols.
- Infra: **Rust, C++, Go, Terraform, OCI, PostgreSQL internals**

Software Development Engineer II Amazon — Seattle, WA & Hyderabad, India Mar 2021 – Oct 2024

- **Real-Time ML Inference Engine:** Architected an **ensemble model pipeline combining XGBoost and Deep Learning** for real-time ads & deal ranking across FreeVe, Twitch, MiniTV, Prime Video, and Amazon Retail — processing millions of events at **100ms inference latency** while driving **5% CTR increase, 30%** higher deal impressions, and **\$5.4M monthly revenue growth**.
- **Distributed Systems & SDK Design:** Designed a uniqueness constraint indexing solution and **SDK** for a multi-tenant NoSQL datastore using **two-phase commit**, achieving **serializable isolation at 3K writes/second** — a developer-facing library used across multiple publisher teams.

- **Conversational AI (BERT):** Developed a **BERT-based inference service** handling **5,000 QPS** at peak, reducing support ticket triage SLA from **7 days** to **2 hours**.
- **Big Data Pipelines:** Engineered **Apache Spark** jobs processing millions of daily events into Amazon Redshift for OLAP with automated dashboards.
- **Infra:** **Python, C++, Kafka, Spark, DynamoDB, Redshift, SageMaker, AWS**

Software Development Engineer Razorpay — Bengaluru, India Aug 2020 – Feb 2021

- Built UPI Payment Gateway Aggregator using **Golang** and Protobuf, processing **1M+ hourly transactions**.
- Maintained **99.99% uptime** with **150ms P99 latency** under **50,000 TPM** peak loads.
- **Infra:** **Go, Kafka, Redis, PostgreSQL, Protobuf**

Software Engineer Mindfire Solutions — Bhubaneswar, India Aug 2018 – Feb 2020

- Developed full-stack workflow tools and an **AR-based product search microservice**, serving 10,000+ daily queries.
- Built administrative dashboards reducing data retrieval latency from 5 min to 1 min.
- **Infra:** **Java, Python, React, MongoDB**

SKILLS

Languages: Rust, C++, Go, Python, Java, Kotlin, C, TypeScript

Systems Programming & Performance: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, eBPF, Kernel-Level Networking, Performance Engineering, Low-Latency Systems

Distributed Systems & Streaming: Apache Kafka, Apache Flink, Apache Spark, Distributed Systems (consensus, replication, sharding), Real-Time Data Pipelines, gRPC, REST API Design

ML Inference & AI: PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), ONNX Runtime, On-Device/Edge Inference, Model Pruning & Quantization, Computer Vision (YOLOv8/v11), Recommendation Systems, MLOps

Cloud & Infra: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Kubernetes (EKS, AKS), Docker, Terraform, Cilium, Kubebuilder, Istio

Data & Storage: DynamoDB, PostgreSQL, Redis, MongoDB, Amazon Redshift, Amazon S3

Core CS: Data Structures & Algorithms, System Design, Operating Systems, Networking (TCP/IP, UDP, TLS, DNS, HTTP/3)

EDUCATION

Bachelor of Technology — Computer Science & Engineering

RCC Institute of Information Technology — Kolkata, India | 2018